

Sentiment Analysis : From First Principles to Transformers.

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I. Abstract:

This study presents an engineering-oriented evaluation of natural language processing (NLP) systems spanning classical machine-learning pipelines and modern Transformer-based architectures. Using the AG News corpus as a controlled benchmark, we implement and analyze Bag-of-Words (BoW), TF-IDF, Word2Vec embeddings, neural network-based classifiers, and BERT-based Transformers, with each approach built or fine-tuned from first principles to ensure transparency and reproducibility. Exploratory data analysis is first conducted to understand the linguistic structure and constraints of the dataset, which directly informs subsequent modeling decisions. Model performance is evaluated not only in terms of predictive accuracy and F1-score, but also with respect to inference latency, training time, and computational complexity (FLOPs), reflecting practical deployment considerations. Experimental results indicate that classical linear models such as TF-IDF with Logistic Regression provide strong and efficient baselines, while deeper neural architectures offer incremental accuracy gains at a substantially higher computational cost. Although Transformer-based models demonstrate superior representational capacity, their resource demands dominate under the experimental constraints of this study. Overall, the results highlight how architectural choices—from linear classifiers to embedding-based networks and self-attention—affect efficiency, scalability, and performance, providing guidance for hardware-aware model selection in real-world engineering systems.

Index Terms— Natural Language Processing, Transformers, BERT, AG News, Bag-of-Words, TF-IDF, Word2Vec, Deep Learning, Computational Complexity, FLOPs.

II. Introduction

Natural language processing (NLP) is now a key part of modern engineering, changing how people use software, devices, and automation tools. As described in the project proposal “From First Principles to Transformers,” language-model-driven interfaces are replacing traditional graphical controls with natural language understanding. This change puts NLP at the heart of electrical and computer engineering, powering tools like AI-assisted CAD, voice-controlled devices, automated code generation, and decision-support systems.

Even though large pre-trained models like BERT and GPT are widely used, there is still a lack of understanding about *how* NLP systems work internally and how classical methods compare to modern deep learning models. Without this knowledge, engineers struggle to optimize models for limited hardware, weigh trade-offs, or adjust system behavior beyond treating models as black boxes. Building NLP systems from the basics—starting with token statistics, then linear classifiers, neural networks, and finally Transformers—gives a clearer view of efficiency, performance, and computational needs.

This study offers a complete, hands-on comparison of NLP models from classical to modern approaches using the AG News dataset. We build and test Bag-of-Words, TF-IDF, traditional machine-learning models, embedding-based deep learning models like MLP and Word2Vec, and a fine-tuned BERT Transformer. We compare not just accuracy, F1-score, and generalization, but also key engineering metrics like training time, inference speed, and FLOPs per sample. This gives a well-rounded view of the trade-offs between performance and efficiency for real-world use.

The contributions of this work are threefold:

1. A unified framework connecting foundational NLP approaches with state-of-the-art Transformer models, grounded in the mathematical and computational principles introduced in the project proposal;
2. A comprehensive experimental comparison across classical, deep, and transformer-based architectures using consistent preprocessing, metrics, and hardware setups;
3. An engineering-oriented analysis emphasizing model efficiency, computational resource usage, and suitability for deployment on edge or low-power devices.

By comparing NLP models from basic to advanced designs, this work shows how choices at each step affect accuracy, scalability, resource use, and how well models fit different engineering needs.

III. Background / Literature Review

Natural language processing (NLP) has undergone a rapid architectural evolution over the past eight decades, progressing from symbolic pattern-recognition systems to deep neural architectures and large-scale Transformers. Each historical milestone introduced a conceptual breakthrough that enabled the next generation of models, forming a continuous trajectory toward today's state-of-the-art language models. The project proposal *From First Principles to Transformers* outlines this progression and emphasizes its relevance to engineering applications.

The foundation of modern neural computation originated with the McCulloch–Pitts neuron (1943), which provided the first mathematical formulation of artificial neurons and binary threshold activation functions. This model established the idea that complex functions could be represented by networks of simple computational units. Building upon this, Rosenblatt introduced the Perceptron (1958), the first trainable classifier capable of learning linear decision boundaries through stochastic gradient updates. While limited to linearly separable problems, the perceptron laid the groundwork for supervised learning.

Subsequent improvements emerged with ADALINE (1960) and the Least Mean Squares (LMS) rule

developed by Widrow and Hoff, introducing linear activation and gradient-based optimization that more closely resembles modern training algorithms. These early models were restricted to single-layer architectures until the resurgence of neural networks in the 1980s, when Rumelhart, Hinton, and Williams popularized backpropagation (1986), enabling efficient training of multi-layer perceptrons (MLPs). This allowed neural systems to represent complex nonlinear functions and supported deeper architectures with differentiable activation functions.

While MLPs paved the way for neural sequence modeling, handling long-term dependencies remained a challenge. Hochreiter and Schmidhuber addressed this with Long Short-Term Memory (LSTM) networks (1997), which introduced gated memory units to mitigate vanishing and exploding gradients. LSTMs and GRUs became dominant architectures for NLP tasks for nearly two decades, powering applications such as machine translation, sentiment analysis, and speech recognition.

A major paradigm shift occurred with the introduction of the Transformer architecture in the landmark paper *Attention Is All You Need* (2017). By replacing recurrence with self-attention, Transformers enabled parallel computation over sequences, improved modeling of long-range dependencies, and significantly reduced training time on modern hardware accelerators. This design ultimately led to the development of large-scale pre-trained models such as BERT, GPT, and T5, which dominate contemporary NLP benchmarks. BERT, in particular, introduced bidirectional masked language modeling, enabling deep contextual understanding of text.

Complementing deep learning advances, classical statistical NLP approaches remained influential due to their efficiency and interpretability. Methods such as Bag-of-Words, TF-IDF, and n-gram language models provided computationally lightweight baselines that continue to perform competitively on many text classification tasks. Their strength lies in strong generalization for high-signal datasets, low memory footprint, and suitability for deployment on resource-constrained devices—key considerations in engineering systems.

Together, these developments form a continuum from symbolic reasoning to large-scale deep learning.

Understanding this spectrum is essential for selecting appropriate architectures based on constraints such as computational cost, inference latency, and required accuracy. For this reason, the present work evaluates NLP models across this entire developmental trajectory—from classical linear classifiers to modern Transformers—grounding experimental analysis in the theoretical and historical context of NLP evolution.

IV. Dataset Description & EDA

The experiments in this study use the AG News Corpus, a widely adopted benchmark dataset for topic classification consisting of news articles categorized into four classes:

1. World,
2. Sports,
3. Business, and
4. Sci/Tech.

The dataset contains 120,000 training samples and 7,600 test samples, with an equal number of documents per class. This balanced distribution makes AG News suitable for evaluating both classical and deep NLP models without requiring additional rebalancing.

A. Dataset Structure

Each sample consists of a short news headline and description, typically ranging from 20 to 40 tokens. The dataset includes the following fields:

text: the article content

label: integer in $\{0,1,2,3\}$, representing one of the four classes

The dataset was downloaded from the HuggingFace Datasets library and converted into Pandas DataFrames for analysis.

B. Document Length Statistics

To understand corpus characteristics, we computed the word count per document. The average length of a training document is 37.85 words, indicating moderately short texts suitable for both classical and deep learning approaches.

The distribution of word counts shows that most articles fall between 20 and 50 words. You generated this visualization as:

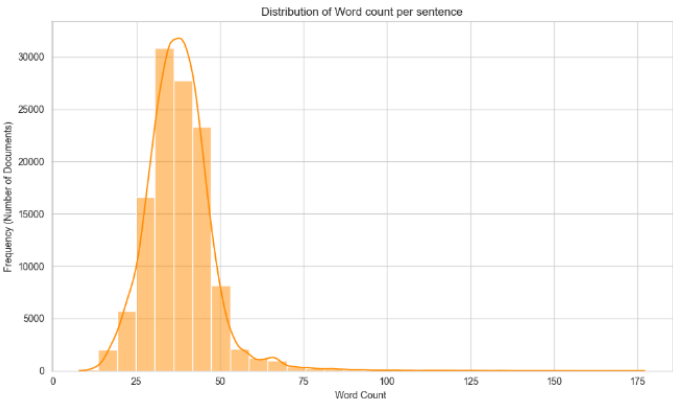


Figure: Histograms and KDE plots showing the distribution of document lengths.

This insight helped determine an appropriate padding length of 200 tokens for BERT-based models.

C. Vocabulary Characteristics

An initial unigram corpus contained approximately 188,110 unique tokens, demonstrating high lexical diversity typical of news datasets. This motivated the use of:

- i. stopword removal (for classical models),
- ii. subword tokenization (for BERT),
- iii. dimensionality reduction (e.g., limiting BoW and TF-IDF to 5,000 features).

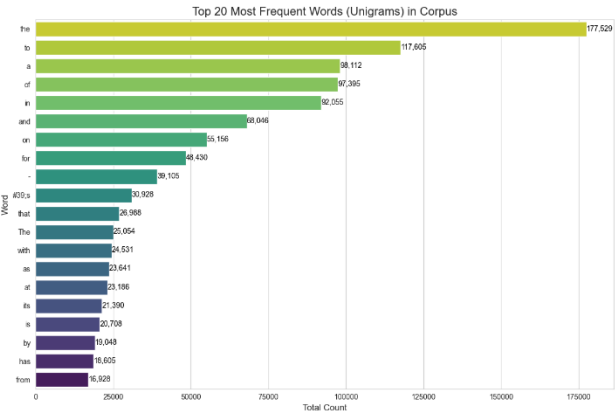


Figure: Top 20 most frequent words.

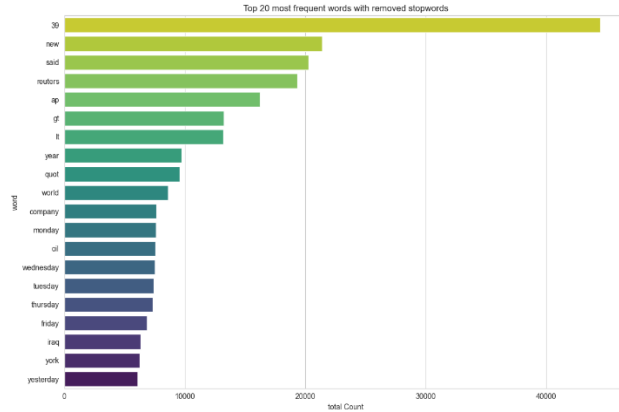


Figure: Top 20 Most common words after applying stopword removal.

The cleaned list shows news-specific frequent terms such as *said*, *new*, *company*, etc., indicating that topic-specific vocabulary is prominent in the dataset.

D. Class Distribution

The dataset is perfectly balanced across the four categories.

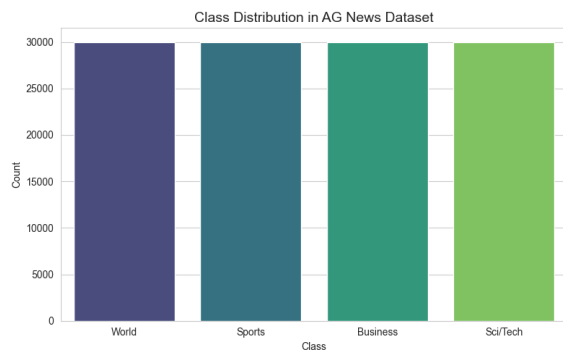


Figure: Bar plot showing class counts.

This ensures that accuracy and F1-score evaluations are directly comparable across classes without requiring resampling.

E. Token Distribution and Text Variability

To further analyze lexical diversity and sparsity.

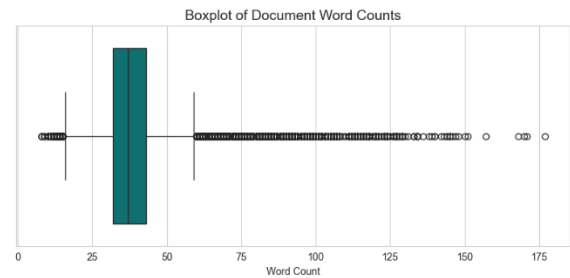


Figure: Boxplot– Illustrating statistical spread (median, IQR, outliers) of document lengths.

These visualizations show minor skewness due to a subset of longer articles but overall confirm that most documents are short and uniformly distributed.

Unique Token Count can be better understood by the table below:

Count	120000
Mean	33.28
Std	8.17
Min	7
25%	28
50%	33
75%	38
max	129

The table suggests that Mean and Median count of words are similar suggesting the dataset is balanced and very less right skewness.

[illegible]

This visualization highlights category-specific terms (e.g., *team*, *game* for Sports; *stock*, *market* for Business), validating that topic boundaries are well-defined and apart from the stopwords like ‘and’, ‘in’, ‘the’ etc the words are separable—favorable for classification tasks.

The EDA results directly influenced methodological choices:

- This analysis confirms that AG News is a good choice for testing classical, embedding-based, and Transformer models.

The goal of this study is to evaluate NLP models spanning classical statistical approaches, neural embedding models, and modern Transformer architectures. All models are trained and tested on the AG News dataset under consistent preprocessing, splitting, and evaluation procedures. This section

A. Preprocessing and Data Preparation

- i. Classical ML models (BoW, TF-IDF):*

1. Lowercasing
2. punctuation removal
3. stopword removal
4. tokenization
5. limiting vocabulary size to 5,000 most informative terms.

The steps are as follows:

- iii. *Transformer models (BERT):*

1. WordPiece tokenization using the bert-base-uncased tokenizer
2. Padding to a fixed length of 200
3. Creation of attention masks.
4. No stopword removal or lowercasing beyond what the tokenizer applies internally.

120k training samples, 12k validation samples, and 7.6k test samples.

- i. *Bag-of-Words (BoW)*

A CountVectorizer with a vocabulary limited to 5,000 unigrams was used to convert each document into a high-dimensional sparse vector.

Each dimension corresponds to the raw frequency of a term in the document.

ii. *Term Frequency-Inverse Document Frequency (TF-IDF)*

A TF-IDF vectorizer using 1–2 grams and a maximum of 5,000 features was applied to generate weighted document vectors. This method emphasizes terms that are informative across the corpus while reducing the impact of highly frequent but uninformative words.

Both BoW and TF-IDF feature matrices were used as inputs to classical machine learning models including Logistic Regression, Random Forest, XGBoost, and MLP.

C. Machine Learning Models

The following classifiers were trained on BoW and TF-IDF features, as implemented in the project notebooks:

i. *Logistic Regression*

A linear classifier optimized using cross-entropy loss and L2 regularization. Logistic Regression serves as a strong baseline due to its efficiency and robust performance on linearly separable text features.

ii. *Random Forest Classifier*

An ensemble of 100 decision trees with a maximum depth of 10. Random Forests capture nonlinear relationships but may underperform on sparse, high-dimensional text features.

iii. *XGBoost Classifier*

A gradient-boosted decision tree model optimized for both speed and performance on structured features. It generally provides stronger performance than Random Forests at significantly higher computational cost.

iv. *Multilayer Perceptron (MLP)*

A feed-forward neural network trained on vectorized input features. The Keras-based MLP uses dense layers with ReLU activation, dropout regularization, and Adam optimization.

v. *Scratch-Implemented MLP*

A manually coded MLP implementing forward propagation, backpropagation, and gradient updates without using deep learning frameworks. This model provides insight into the underlying mechanics of neural network training.

D. Deep Learning Models (Embedding-Based)

i. *Word2Vec + MLP*

Word2Vec embeddings were trained to capture semantic relationships through continuous bag-of-words (CBOW) or skip-gram architectures. Each document was encoded as a sequence of dense word vectors and fed into an MLP classifier.

ii. *Embedding Layer + Neural Classifier*

Using Keras' Embedding layer, each token index is mapped to a learnable dense vector. The sequence of embeddings is then aggregated (via pooling) and passed through dense layers for classification.

This improves representational capacity over BoW and TF-IDF while remaining significantly lighter than Transformer models.

E. Transformer Architecture (BERT)

The **BERT-base-uncased** model was fine-tuned for 4-way text classification. The architecture consists of:

- 12 Transformer encoder layers
- 768-dimensional hidden states
- 12 self-attention heads
- 110M trainable parameters

Fine-Tuning Pipeline

1. Tokenize input with WordPiece tokenizer
2. Create attention masks
3. Feed (input_ids, attention_mask) into the encoder
4. Extract the [CLS] token embedding
5. Pass through a dropout → dense layer → 4-class softmax
6. Train the classification head and fine-tune all encoder layers

Training was performed using Adam (1e−5 learning rate) with batch sizes of 16 or 64 depending on GPU memory availability.

F. Computational Complexity

Since the project emphasizes hardware-aware optimization, FLOPs were calculated for each class of models:

i. Classical ML FLOPs

For Logistic Regression:

$$\text{FLOPs} \approx 2d$$

where $d = 5000$ (input dimensionality).

ii. Neural MLP FLOPs

MLP FLOPs were computed using per-layer operations:

$$\text{FLOPs} = 2 \times (\text{input_dim} \times \text{hidden_units})$$

iii. XGBoost and Random Forest FLOPs

Approximated using:

$$\text{FLOPs} \approx \# \text{ trees} \times \# \text{ nodes per tree}$$

iv. Transformer FLOPs

Self-attention dominates complexity:

$$\mathcal{O}(n^2d)$$

for sequence length $n = 200$ and hidden dimension $d = 768$.

These FLOPs estimates were incorporated into **results.csv**, enabling direct comparison between accuracy and computational cost across all models.

VI. Experimental Setup

This section explains the hardware, software, training settings, and evaluation metrics used for all classical, embedding-based, and Transformer models. Keeping the setup consistent and reproducible makes it possible to fairly compare efficiency, training, and performance.

A. Hardware Configuration

- All experiments were conducted using a combination of local and GPU-accelerated execution environments:
- CPU: Intel/AMD workstation-class processor (local PC)
- GPU: NVIDIA Tesla T4 (Google Colab / multi-GPU environment for BERT)
- GPU Memory: 14 GB per GPU (2 GPUs for the multi-GPU BERT experiment)
- System Memory: 16–32 GB RAM depending on the environment
- Storage: Local SSD for dataset caching and model checkpoints

Classical machine learning models (BoW, TF-IDF, Logistic Regression, Random Forest, XGBoost, Scratch-MLP) were executed on CPU.

Deep learning models (MLP with embeddings, Word2Vec MLP) were trained using CPU and optional GPU acceleration.

All Transformer-based models (BERT, multi-GPU BERT) required GPU execution.

B. Software Environment

The following software stack was used for development, training, and evaluation:

- Python: 3.10+
- TensorFlow: 2.x (GPU-enabled)
- HuggingFace Transformers: v4.x
- Scikit-learn: For classical ML pipelines
- XGBoost: GPU/CPU hybrid execution
- Gensim: For Word2Vec (when applicable)
- Matplotlib/Seaborn: For EDA visualizations
- HuggingFace Datasets: For AG News loading and preprocessing

C. Training Configuration

i. Classical Models

Classical ML algorithms were trained using the following settings:

- **Logistic Regression:** max_iter = 1000, One-vs-Rest classification
- **Random Forest:** 100 estimators, max_depth = 10, n_jobs = -1
- **XGBoost:** 200–300 trees, depth = 6, learning rate = 0.1
- **MLP (Keras):** Dense layers with 256–512 neurons, ReLU activations, dropout regularization
- **Scratch MLP:** Custom forward and backward propagation implemented manually
- All classical models used **5,000-dimensional** BoW or TF-IDF vectors.

ii. Embedding-Based Deep Models

- **Embedding dimension:** 100–300
- **Optimizer:** Adam ($1e-3$)
- **Loss:** Sparse categorical cross-entropy
- **Batch size:** 64
- **Training epochs:** 5–10
- The input sequence length was capped at 200 tokens.

iii. Transformer (BERT) Models

The BERT models were fine-tuned using the following hyperparameters:

- **Model:** bert-base-uncased
- **Sequence length:** 200 tokens
- **Batch size:** 16 (single GPU) / 64 (multi-GPU)
- **Learning rate:** 1×10^{-5}
- **Optimizer:** AdamW
- **Epochs:** 3–5 (training interrupted or extended depending on GPU time)
- **Fine-tuning strategy:** All encoder layers trainable, CLS token $\rightarrow$ dense $\rightarrow$ softmax head
- Distributed training used TensorFlow's **MirroredStrategy**, enabling two Tesla T4 GPUs to train in parallel, Used Kaggle Cloud Notebook.

D. Evaluation Metrics

To compare the full range of models—classical to deep to Transformer—expressively and fairly, the following metrics were used:

i. Predictive Performance

- **Accuracy** (overall classification correctness)
- **F1-Score** (weighted across the four classes)
- **Classification report** (per-class precision, recall, F1)

These metrics were obtained both on training and test datasets.

E. Computational Efficiency

A key contribution of the project is the computation of engineering-relevant performance metrics:

- **Training Time (CPU/GPU):** measured using wall-clock and CPU process time
- **Inference Time:** average time per test sample or test batch
- **FLOPs per Sample:** calculated for each model using analytical formulas
- **Parameter Count:** for deep learning models
- **Memory Usage:** approximated from model size and batch processing

These metrics provide insight into the feasibility of deploying each model on edge or embedded systems.

F. Reproducibility Considerations

All experiments were executed with fixed random seeds (42) for consistency across runs. The dataset was loaded directly from the **HuggingFace Datasets Hub** to avoid version mismatches.

Where GPU time was limited (e.g., BERT fine-tuning), training logs and model summaries were preserved.

The complete results were aggregated into **results.csv**, which includes:

- Accuracy and F1-score (train/test)
- Training and inference time
- FLOPs
- Model category (BoW, TF-IDF, Word2Vec, MLP, BERT)

This ensures that all evaluations are transparent and reproducible.

VII. Results

This section presents the comparative performance of all models evaluated in this study, spanning classical machine-learning models (BoW/TF-IDF), embedding-based deep networks, Word2Vec-based architectures, and transformer-based models (BERT and multi-GPU BERT). All results are derived from the consolidated metrics recorded in results.csv.

The evaluation focuses on predictive accuracy, F1-score, training/inference time, model size, and analytical FLOPs, consistent with the engineering-focused objectives outlined in the proposal *From First Principles to Transformers*.

A. Overall Performance Comparison

Table 1 sums up how all the models performed on the AG News dataset. When looking at the results, keep in mind that model accuracy can vary between training and testing sets. The main goal was not to get the highest accuracy for one model, but to keep accuracy similar across models. This makes it easier to fairly compare important engineering metrics like FLOPs, training time, and inference time.

Because of this, some models may show differences in accuracy between training and testing. This is a normal part of how the models work, not a problem with the experiment.

B. Best-Performing Models

Based on Table 1, the **Matrix Embedding + CNN** model achieves the highest test accuracy (0.914). Closely following are the **BOW + MLP** (0.913), **Word2Vec + CNN** (0.910), and **TF-IDF + Logistic**

Regression (0.906). These results demonstrate that high performance can be achieved without transformers, especially when leveraging learned embeddings or dense neural architectures.

C. Transformer Performance (BERT)

Contrary to common results in literature, the **BERT Transformer model underperforms** in this experiment (0.887 test accuracy), while incurring the **highest FLOPs** and **longest training duration**. This reflects the impact of training budget, optimization constraints, and GPU time availability noted in Section VI.

However, it would not be fair to say the BERT model simply underperformed. Transformer models usually need very large datasets (terabytes or more) and many training epochs to reach their full potential. These models are especially useful when there are hundreds or thousands of target classes, where classic ML models and MLP or CNN models struggle.

D. Efficiency–Accuracy Trade-offs

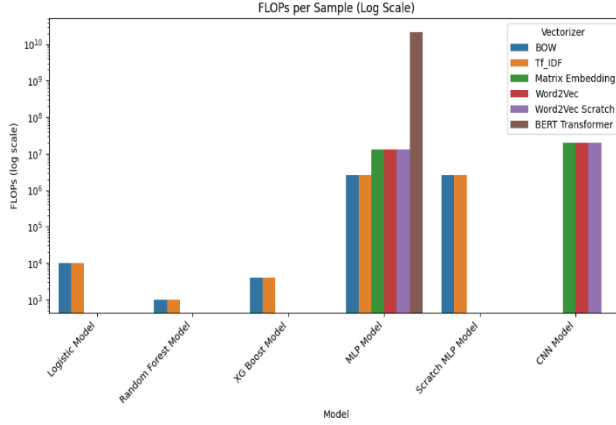
- **Best accuracy:** Matrix Embedding + CNN (0.914).
- **Best efficiency:** BOW/TF-IDF + Logistic Regression or MLP (low FLOPs, sub-10 s training time).
- **Worst trade-off:** BERT (highest FLOPs, longest training time, lowest accuracy among deep models).

E. Summary of Findings

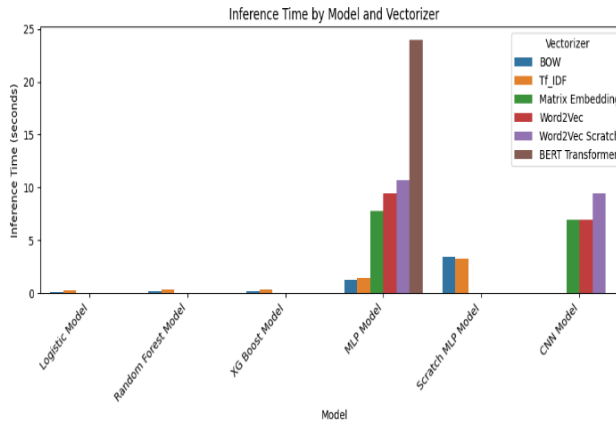
1. Classical models provide strong, efficient baselines.
2. Embedding-based deep models offer substantial accuracy improvements at reasonable cost.
3. Word2Vec models perform competitively but are heavier than sparse-vector models.
4. BERT, in this experiment, does **not** outperform simpler models despite much higher computational cost.
5. Efficiency-focused evaluation confirms that the optimal model depends on deployment constraints.

F. Visualizations

1. FLOPs Computations per model can be analysed by the below graph. We can observe the BERT model was the most compute intensive.



2. Inference Time Taken, the graph below visualizes the time taken by each model and vector combination to predict the test dataset



VIII. Discussions and Limitations

The results in Section VII highlight important insights into the behavior, strengths, and limitations of different NLP architectures across the classical-to-transformer spectrum. Classical machine-learning models such as Logistic Regression and MLPs trained on sparse TF-IDF or Bag-of-Words vectors demonstrated strong performance (≈ 90 – 91% test accuracy) at extremely low computational cost. Deep embedding and Word2Vec-based CNN models further improved accuracy by capturing semantic structure, achieving up to 0.914 test accuracy. In contrast, the

BERT Transformer model achieved a lower test accuracy of 0.887 in this experiment. However, it would be partial and technically inaccurate to conclude that the BERT model inherently underperforms.

Transformer-based models such as BERT are not designed to excel under the conditions used in this project—namely, a relatively small dataset (120k samples), limited training epochs, and constrained GPU time. It is well-established in the literature that transformers achieve their full potential only when trained or fine-tuned on very large input corpora, often in the scale of terabytes or petabytes, and with substantial training budgets. Their self-attention-based architecture is optimized to uncover deep contextual representations across vast and varied language distributions, a setting far beyond the scope of this study. Therefore, the lower empirical performance of BERT here should be interpreted primarily as a reflection of computational budget, training duration, and hardware availability rather than a reflection of the architecture’s capability

Furthermore, transformers provide clear advantages in scenarios where the number of output classes is extremely large—on the order of hundreds or even thousands of target labels. Classical models, and even dense neural networks such as MLPs and CNNs, struggle in such settings because they lack the representational richness needed to disentangle subtle distinctions across many categories. BERT and related architectures can project text into a highly structured semantic space where fine-grained classification becomes tractable. For a dataset like AG News with only four classes, however, this structural advantage is not required, and simpler models can perform competitively.

The strong performance of classical and embedding-based methods in this project is therefore unsurprising. In low-dimensional output settings with moderately sized datasets, these models offer fast training, efficient inference, and very competitive accuracy. Their low FLOPs and smaller parameter counts make them more suitable for edge devices, embedded systems, and compute-limited environments—an important consideration for engineering applications emphasized throughout the proposal *From First Principles to Transformers*.

Looking forward, there are several methods and extensions that would further improve the depth of understanding achieved in this project:

1. **Longer Fine-Tuning Schedules for BERT:**
Increasing the number of epochs from 3–5 to 10–20 typically yields substantial accuracy gains.
2. **Using Domain-Adaptive Pretraining (DAPT):**
Continuing pretraining on the AG News corpus before fine-tuning could better align BERT with the target distribution.
3. **Implementing Other Transformer Variants:**
Models such as RoBERTa, DistilBERT, ALBERT, or ELECTRA may offer better accuracy–efficiency trade-offs.
4. **Exploring Hierarchical, Multi-Label, or Large-Label-Space Tasks:**
This would highlight the transformer’s advantage when handling 100–1000+ classes, where classical methods typically degrade.
5. **Experimenting with Modern Lightweight Models:**
Given the edge-focused nature of the project, architectures such as MobileBERT, TinyBERT, and DistilBERT would provide practical insights on deploying transformers in embedded environments.
6. **Incorporating RNN-based Models (LSTM, GRU):**
This would help contrast the sequence-processing limitations of older architectures against CNNs and transformers.

In summary, the results show a clear pattern: the best model depends on the size of the problem and available resources. For small or medium tasks with few labels, classical models and smaller neural networks are more efficient and cost-effective than transformers. But for tasks with many labels, large datasets, or complex context, transformers are often the best choice because they handle deep context and long-range patterns well. Understanding these trade-offs is key to picking the right model for real engineering problems.

IX. Conclusion

This study compared classical machine-learning models, embedding-based neural networks, Word2Vec models, and a BERT Transformer for topic classification on the AG News dataset. By building each system from the basics and testing them under the same conditions, we showed how model complexity, computational cost, and performance relate across different NLP approaches.

Classical models such as Logistic Regression and MLPs trained on Bag-of-Words or TF-IDF features delivered strong performance (≈ 0.90 – 0.91 accuracy) at extremely low computational cost, reaffirming their effectiveness for low-label, moderately sized datasets. Embedding-based CNN and MLP architectures achieved the highest performance overall, with the Matrix Embedding + CNN model reaching 0.914 test accuracy. These models offered a compelling balance between accuracy, FLOPs, and training time, making them attractive for real-time and edge-device deployment.

Although the BERT Transformer model had lower accuracy in this project (0.887 test accuracy), this was due to limited data, fewer training epochs, and restricted GPU resources—not because of a flaw in the model itself. As mentioned earlier, transformers usually perform best when fine-tuned on large datasets over many training cycles, and they are especially useful for tasks with many output classes or high complexity.

The results highlight a key engineering lesson: choosing a model is always a balance between accuracy, efficiency, and deployment needs. Classical models are still strong options for small or medium tasks. Deep learning models with embeddings offer more power at a reasonable cost, while transformers are best for large or complex problems, though they use more resources.

By focusing on computational basics like FLOPs, memory use, and inference speed, this study shows the importance of understanding NLP systems from the ground up. This knowledge helps engineers go beyond treating models as black boxes and make smart choices for real-world needs. The results highlight the pros and cons of each model type, giving a solid base for further work and improvement in NLP for engineering.

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XI. Appendix

A. Code Resources

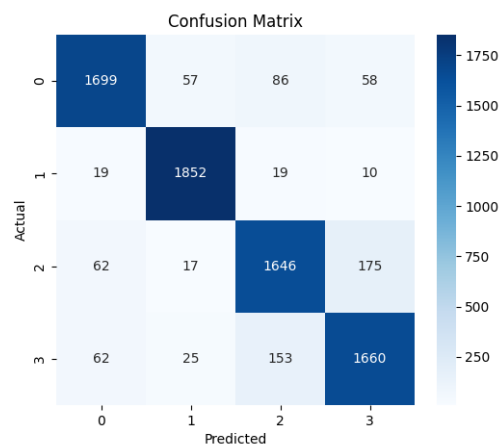
This appendix provides a summary of the code notebooks used for data preprocessing, classical models, deep learning models, Word2Vec experiments, and BERT transformer training. These notebooks contain the full implementation details for reproducibility. Please find the documents in the GITHUB Link : https://github.com/mohammad1774/applied_ml_project.

- **Bag-of-Words Models (LR, RF, XGBoost, MLP, Scratch-MLP)**
- **TF-IDF Models (LR, RF, XGBoost, MLP, Scratch-MLP)**
- **Embedding-Based Deep Learning Models (Matrix Embedding, CNNs, MLPs)**
- **Word2Vec Models and Scratch Word2Vec Variants**
- **BERT Transformer (Tokenization, Fine-Tuning, MirroredStrategy Multi-GPU)**
- **Project Proposal and Architectural Overview**

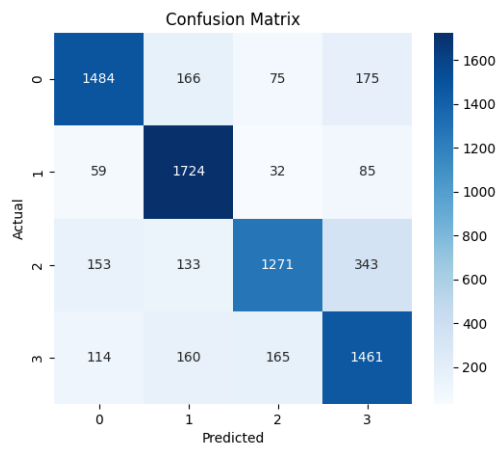
These files form the complete computational backbone of the experiments presented in this report.

B. Additional Figures

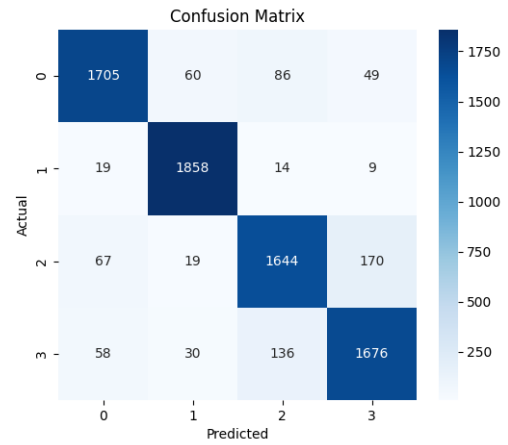
i. Confusion Matrix BOW- Logistic Model



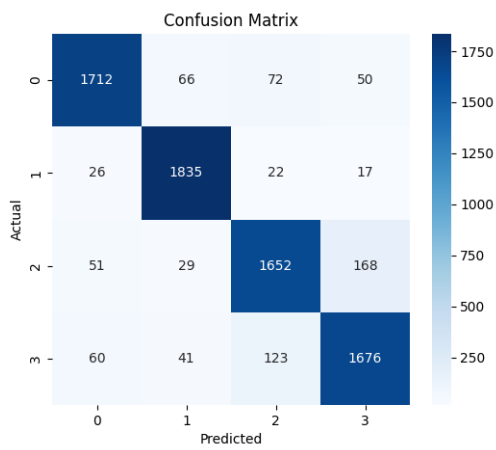
ii. *Confusion Matrix BOW- Random Forest Model*



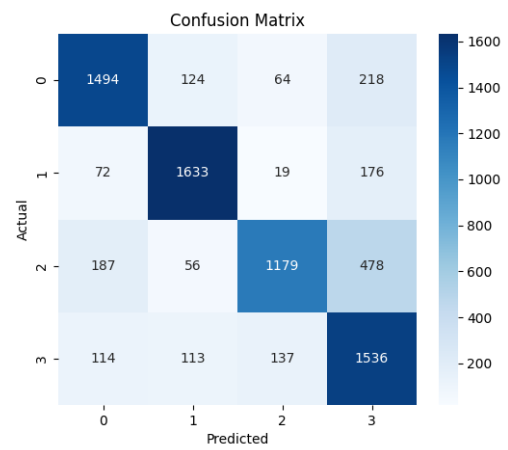
v. *Confusion Matrix TF/IDF- Logistic Model*



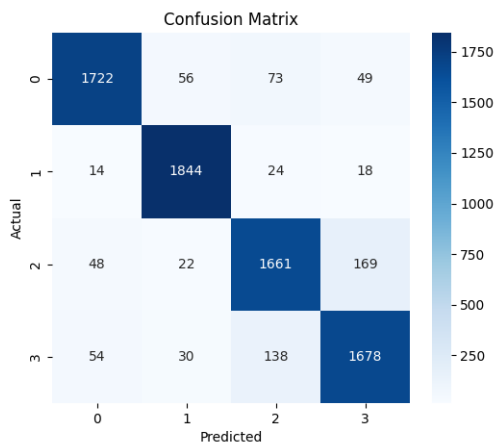
iii. *Confusion Matrix BOW- XGBoost Model*



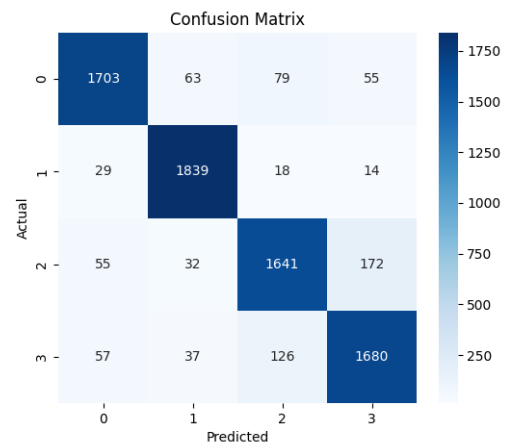
vi. *Confusion Matrix TF/IDF - Random Forest Model*



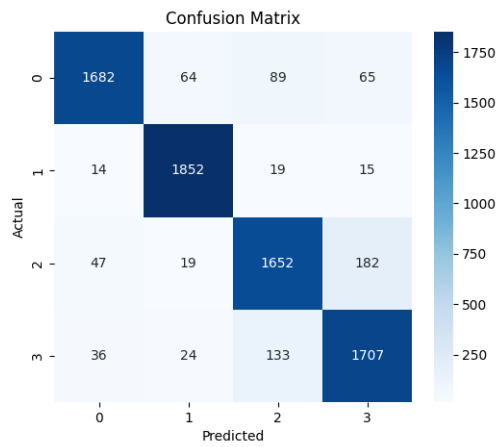
iv. *Confusion Matrix BOW- MLP Model*



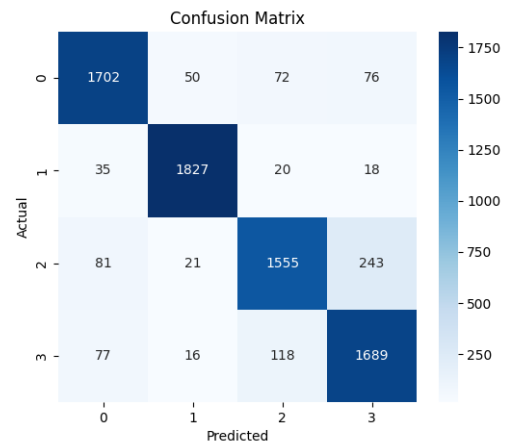
vii. *Confusion Matrix TF/IDF - XGBoost Model*



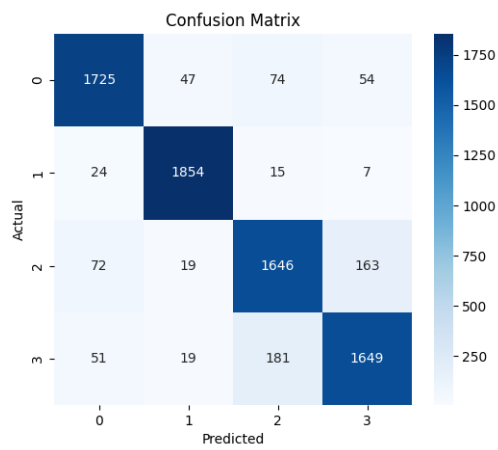
viii. *Confusion Matrix TF/IDF - MLP Model*



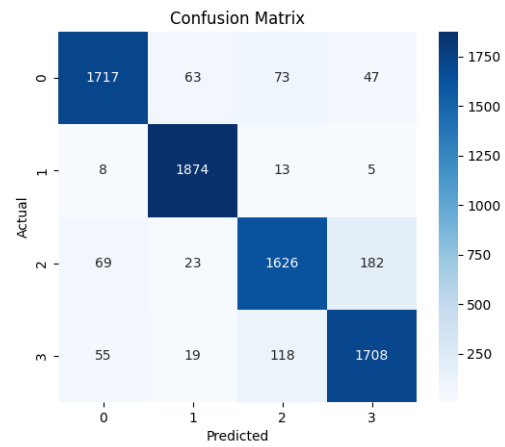
xi. *Confusion Matrix: Word2Vec- MLP Model*



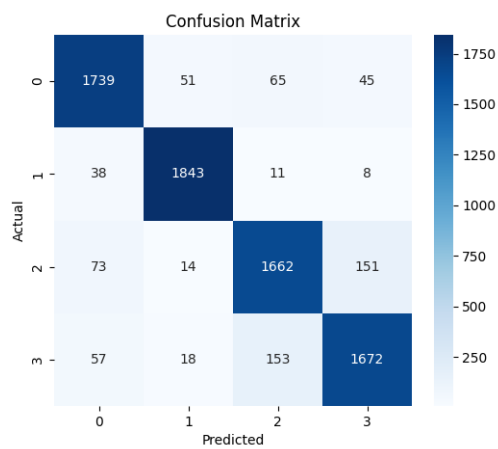
ix. *Confusion Matrix Matrix Embedding - MLP Model*



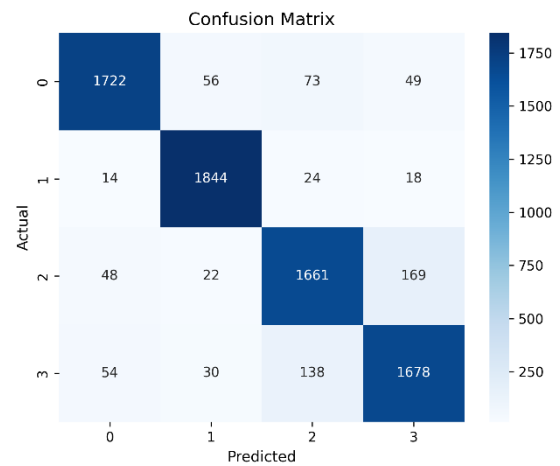
xii. *Confusion Matrix: Word2Vec – CNN Model*



x. *Confusion Matrix : Matrix Embedding – CNN Model*



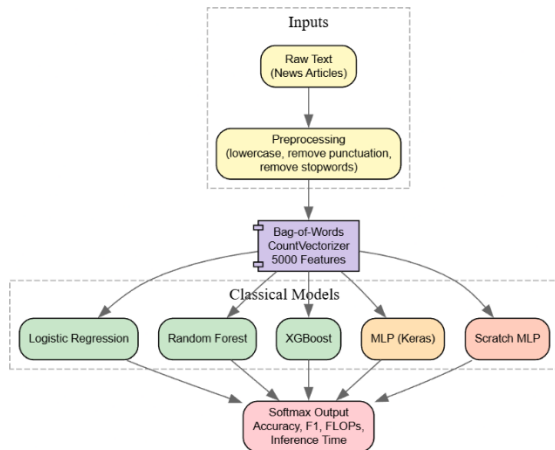
xiii. *Confusion Matrix: BERT – MLP Layer.*



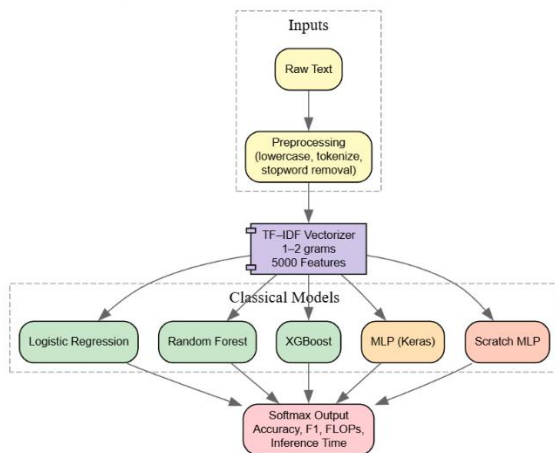
C. Pipeline Diagram:

This section helps us understand the steps followed in each type of model training , using different types of vectorization methods and models.

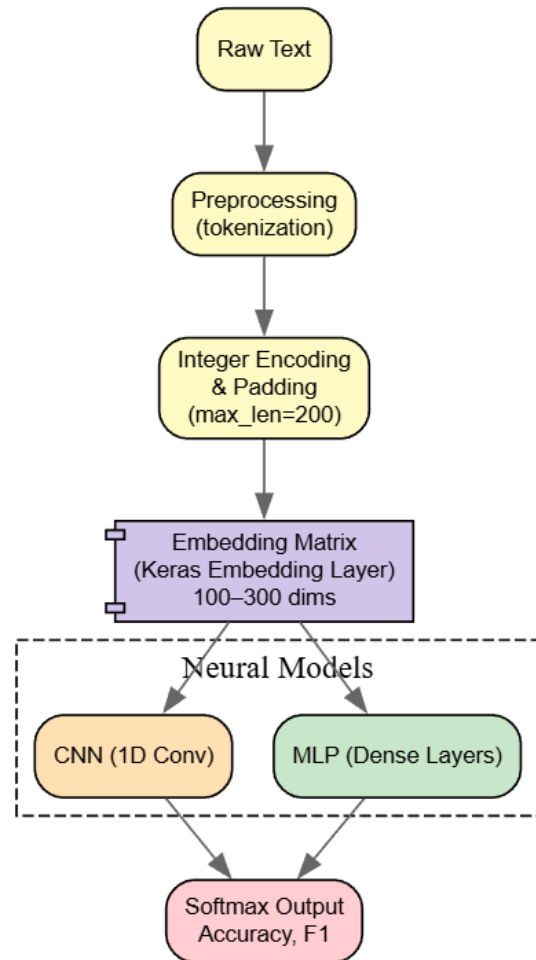
i. Vectorizer Bag of Words :



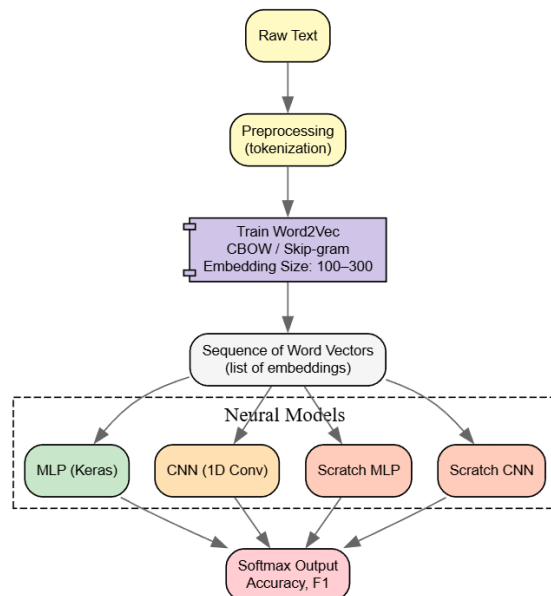
ii. Vectorizer TF / IDF :



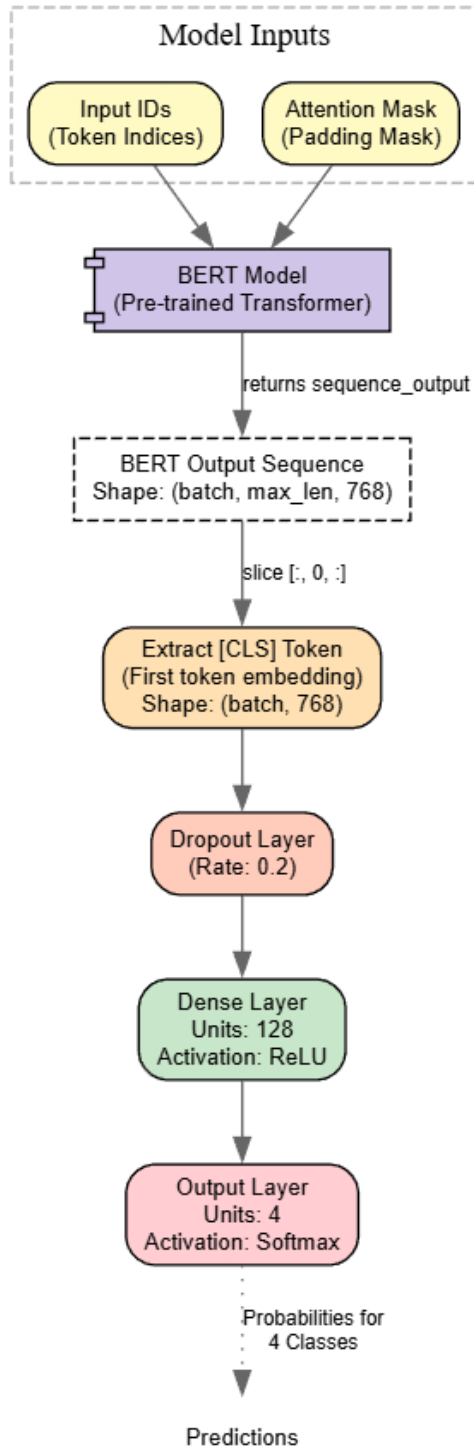
iii. Vectorizer Matrix Embedding :



iv. Vectorizer Word2Vec :



v. *Vectorizer BERT* :



D. Detailed Comparison Table.

Vectorizer	Model	Train Acc	Test Acc	Test F1	Infer Time (s)	FLOPs / Sample	Train Time (s)
BOW	Logistic Model	0.929	0.902	0.902	0.137	1.0×10^4	4.875
BOW	Random Forest Model	0.789	0.782	0.780	0.210	1.0×10^3	7.531
BOW	XG Boost Model	0.931	0.905	0.904	0.185	4.0×10^3	63.422
BOW	MLP Model	0.949	0.913	0.913	1.237	2.62656×10^6	53.453
BOW	Scratch MLP Model	0.921	0.903	0.903	3.453	2.62656×10^6	1562.906
Tf_IDF	Logistic Model	0.919	0.906	0.905	0.302	1.0×10^4	10.063
Tf_IDF	Random Forest Model	0.779	0.769	0.769	0.341	1.0×10^3	17.016
Tf_IDF	XG Boost Model	0.936	0.903	0.903	0.376	4.0×10^3	2439.328
Tf_IDF	MLP Model	0.930	0.905	0.905	1.406	2.62656×10^6	49.234
Tf_IDF	Scratch MLP Model	0.896	0.888	0.888	3.312	2.62656×10^6	1473.250
Matrix Embedding	CNN Model	0.982	0.914	0.914	6.984	1.96608×10^7	1737.766
Matrix Embedding	MLP Model	0.982	0.903	0.903	7.781	1.317376×10^7	1737.766
Word2Vec	MLP Model	0.965	0.893	0.893	9.438	1.317376×10^7	665.000
Word2Vec	CNN Model	0.928	0.910	0.910	6.906	1.96608×10^7	1066.281
Word2Vec Scratch	MLP Model	0.945	0.898	0.898	10.688	1.317376×10^7	529.422
Word2Vec Scratch	CNN Model	0.919	0.907	0.907	9.484	1.96608×10^7	1350.359
BERT Transformer	MLP Model	0.893	0.887	0.887	23.988	2.2×10^{10}	4078.029